Activity 10

1. Hamilton's principle states that the path a particle follows between points 1 and 2 in the time interval t_1 to t_2 is that which renders the action:

$$S = \int_{t_1}^{t_2} \mathcal{L}\, dt = \int_{t_1}^{t_2} T - U\, dt \tag{1}$$

stationary. This is true when the Lagrange equations are satisfied:

$$\frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial q_i'} = 0$$

(a) Show that the Euler Lagrange equations reproduce Newton's 2nd Law in Cartesian coordinates.

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial T}{\partial x} - \frac{\partial U}{\partial x} = 0 - \frac{\partial U}{\partial x} = -(-F_x) = +F_x \qquad \text{Since } \vec{F} = -\vec{\nabla} U = -\frac{\partial U}{\partial x}\hat{x} - \frac{\partial U}{\partial y}\hat{y} - \frac{\partial U}{\partial z}\hat{z}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{\partial T}{\partial \dot{x}} - \frac{\partial U}{\partial \dot{x}} = \frac{\partial}{\partial \dot{x}}\left(\tfrac{1}{2}m\dot{x}^2\right) - 0 = m\dot{x} = mv_x = p_x \qquad \frac{\partial \mathcal{L}}{\partial x} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} \Rightarrow F_x = \frac{d}{dt}(m\dot{x}) = m\ddot{x} = ma_x$$

(continued on next page)

(b) Write down the Lagrange equations for a particle in the 2D plane using polar coordinates moving while being acted on by a conservative force $\vec{F} = (F_r, F_\theta)$. Interpret the meaning of the result you get from the Euler-Lagrange equation. Show that you get Newton's second law again in polar coordinates.

$$\mathcal{L} = T - U = \tfrac{1}{2}m\vec{v}^2 - U$$

Find T: $\vec{v} = \dot{\vec{r}} = \frac{d\vec{r}}{dt}$

$$d\vec{r} = dr\,\hat{r} + r\,d\theta\,\hat{\theta}$$

$$\frac{d\vec{r}}{dt} = \frac{dr}{dt}\hat{r} + r\frac{d\theta}{dt}\hat{\theta} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$$

$$\vec{v}^2 = \dot{r}^2 + r^2\dot{\theta}^2$$

$$\mathcal{L} = \tfrac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - U(r,\theta)$$

$$\frac{\partial \mathcal{L}}{\partial r} = mr\dot{\theta}^2 - \frac{\partial U}{\partial r}$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{r}} = \frac{d}{dt}(m\dot{r}) = m\ddot{r}$$

$$\Rightarrow m\ddot{r} = mr\dot{\theta}^2 - \frac{\partial U}{\partial r}$$

$$(1) \quad m\ddot{r} - mr\dot{\theta}^2 = -\frac{\partial U}{\partial r}$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = -\frac{\partial U}{\partial \theta}$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \frac{d}{dt}(mr^2\dot{\theta}) = 2mr\dot{r}\dot{\theta} + mr^2\ddot{\theta}$$

$$-\frac{\partial U}{\partial \theta} = 2m\dot{r}r\dot{\theta} + mr\ddot{\theta}$$

$$(2) \quad -\frac{1}{r}\frac{\partial U}{\partial \theta} = 2m\dot{r}\dot{\theta} - m\ddot{\theta}$$

[in polar coords, see Ch. 1]

$$ma_r = m(\ddot{r} - r\dot{\theta}^2) = -\partial U/\partial r \quad \text{from (1)}$$

$$ma_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = -\frac{1}{r}\frac{\partial U}{\partial r} \quad \text{from (2)}$$

$$m\vec{a} = ma_r\hat{r} + ma_\theta\hat{\theta} = -\frac{\partial U}{\partial r}\hat{r} - \frac{1}{r}\frac{\partial U}{\partial r}\hat{\theta} = -\vec{\nabla}U \quad \text{in polar coords}$$

$$\Rightarrow m\vec{a} = -\vec{\nabla}U = \vec{F} \Rightarrow \vec{F} = m\vec{a} \quad \boxtimes$$

$$\mathcal{L} = T - U$$

$$\mathcal{L} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - U(x,y,z)$$

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} \qquad \frac{\partial \mathcal{L}}{\partial y} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{y}} \qquad \frac{\partial \mathcal{L}}{\partial z} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{z}}$$

$$F_x = ma_x \qquad F_y = ma_y \qquad F_z = ma_z \quad \Rightarrow \quad \vec{F} = m\vec{a}$$

For each physical system below, write down the Lagrangian $\mathcal{L}$ using the minimum number of generalized coordinates to fully describe the motion of the system, and find the Lagrange equations of motions. Then answer the additional question posed for each system.

1. Consider the Atwood machine shown in Fig. 1, with two masses m_1 and m_2 connected by an inextensible string of length l, which passes over a pulley with frictionless bearings of mass M and radius R. Solve the Lagrange equations of motion to find the acceleration of the masses.

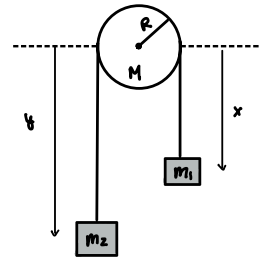

Figure 1: An Atwood machine

2. Consider a block sliding on a wedge, which itself is sliding on a frictionless surface, see Fig. 2.

   (a) How long does it take for the block to reach the bottom of the wedge?

   (b) Check that the acceleration of the block in the limit $M \to 0$ makes sense.

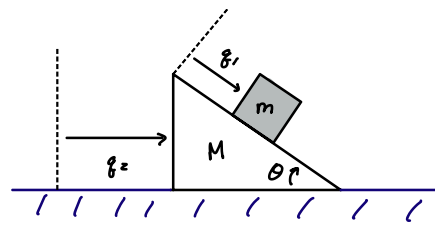

Figure 2: A block sliding on a wedge, which is sliding on a surface. All surfaces are frictionless.

3. Consider a bead threading onto a frictionless wire hoop, as shown in Fig. 3. The hoop rotates about its vertical diameter with a constant angular velocity $\dot{\phi} = \omega$. Find the equilibrium points for the bead, and classify each equilibrium position as stable or unstable.

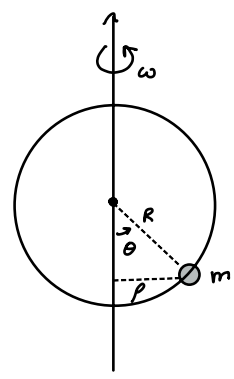

Figure 3: A bead on a spinning hoop, whose position is specified by θ.

4. Consider a bead threaded through a smooth wire that takes the shape of a helix, as shown in Fig. 4. The shape of the helix is given by $\rho = R$ and $z = \lambda\phi$, where ρ, z, ϕ are cylindrical coordinates. Gravity points in the $-\hat{z}$ direction.

   (a) Find the vertical acceleration $\ddot{z}$ for the bead.

   (b) Find $\ddot{z}$ in the limit where $R \to 0$. Does this make sense?

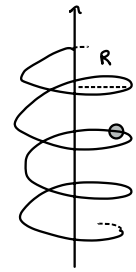

Figure 4: A bead on a wire shaped into a helix

5. Consider the double Atwood machine shown in Fig. 5. Explain why the top pulley rotates even though there are equal masses on both sides. *Hint: Proceed by contradiction. Consider what the tension force would be in the top pulley's string if the top pulley's string was not moving. If that were the correct tension, what would the net forces on the second pulley system be? What would the motion of the CM of the second pulley system be? Does that give a contradiction?*

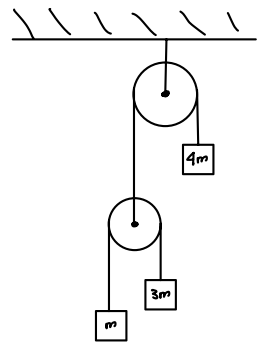

Figure 5: A double Atwood machine.

6. Consider a pendulum attached to a fixed point on a wheel as soon in Fig. 6. The wheel is forced to rotate at a fixed angular velocity ω around its center. Check that the equations of motion make sense when $\omega \to 0$.

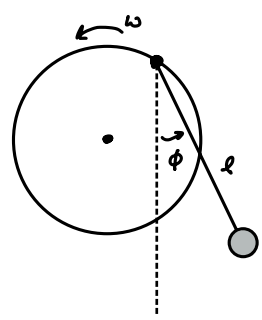

Figure 6: A pendulum attached to a wheel of radius R.

$$\frac{\partial \mathcal{L}}{\partial \phi} = \frac{1}{2} m \left[2 l R \omega \dot{\phi} \cos(\phi - \omega t) \right] - mg \left[l \sin\phi \right]$$

$$= m l R \omega \dot{\phi} \cos(\phi - \omega t) - mgl \sin\phi$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \frac{1}{2} m \left[2 l R \omega \sin(\phi - \omega t) + 2 l^2 \dot{\phi} \right]$$

$$= m l R \omega \sin(\phi - \omega t) + m l^2 \dot{\phi}$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = m l R \omega \cos(\phi - \omega t)(\dot{\phi} - \omega) + m l^2 \ddot{\phi}$$

EOM:

$$\cancel{m} \cancel{l} R \omega \dot{\phi} \cos(\phi - \omega t) - \cancel{m} g \cancel{l} \sin\phi$$

$$= \cancel{m} \cancel{l} R \omega \cos(\phi - \omega t)(\dot{\phi} - \omega) + \cancel{m} l^{\cancel{2}} \ddot{\phi}$$

$$\Rightarrow \cancel{R \omega \dot{\phi} \cos(\phi - \omega t)} - g \sin\phi = R \omega (\cancel{\dot{\phi}} - \omega) \cos(\phi - \omega t) + l \ddot{\phi}$$

$$-g \sin\phi = -R\omega^2 \cos(\phi - \omega t) + l \ddot{\phi}$$

$$\Rightarrow \boxed{\ddot{\phi} = \frac{R\omega^2}{l} \cos(\phi - \omega t) - \frac{g}{l} \sin\phi} \quad \text{EOM}$$

When $\omega \to 0$:

$$\lim_{\omega \to 0} \ddot{\phi} \to -\frac{g}{l} \sin\phi$$, the equation for a normal pendulum.